CHERI FAVEL



Chemical Engineer (MEng) | Water Treatment & Resource Recovery

Abu Dhabi, UAE | UAE resident | Available immediately

cherifavel@gmail.com | +971 58 584 5473 | linkedin.com/in/cheri-favel | Engineering portfolio

PROFESSIONAL PROFILE

MEng Chemical Engineer with industrial R&D experience in mine-tailings water treatment, dissolved-metal removal and electrochemical resource recovery. Experience includes treatment testwork, analytical data interpretation, mass and energy balances, flow-diagram preparation and technical reporting. Additional hands-on support with vessel RO watermakers, pumps and plumbing, supported by environmental engineering training and published mineral-leaching research. Seeking UAE water-treatment, environmental process, applications or R&D opportunities.

CORE COMPETENCIES

- **Industrial water treatment**: mine-tailings water, iron/zinc/manganese removal, oxidation, precipitation, electrochemical treatment, process evaluation and resource recovery.
- **Water chemistry / separation**: pH adjustment, metal speciation considerations, solid–liquid separation, sampling and analysis; understanding of UF/NF/RO principles and fouling/scaling.
- **Process engineering**: mass/energy balances, process calculations, PFDs/P&IDs, equipment-sizing support, design bases, specifications and technical reporting.
- **Laboratory / analytical**: ICP-OES, XRF, XRD, ATR-FTIR, SEM-EDS, BET, TGA, pH and particle-size analysis; experimental design, data interpretation, kinetic modelling and test documentation.
- **Software / equipment**: advanced Excel (VBA, Power Query, PivotTables), Visio and Python; foundational Aspen Plus/HYSYS exposure; vessel RO watermaker and pump maintenance support.

ENGINEERING EXPERIENCE

Process & R&D Engineer
Mar 2021 – Jul 2022

Free Radical Process Design (FRPD) | South Africa

- Conducted laboratory investigations into industrial water-treatment and hydrometallurgical processes; evaluated chemical and electrochemical treatment routes; prepared mass and energy balances, process calculations, flow diagrams (BFDs/PFDs) and reports; reviewed and revised design bases; supported equipment sizing.
- Managed laboratory investigations, interpreted analytical results and evaluated process feasibility; prepared engineering recommendations and reviewed supplier quotations.

Postgraduate Researcher in Chemical Engineering
Jan 2020 – Dec 2021

University of Pretoria | South Africa

- Designed mineral-leaching experiments and identified an internal diffusion-controlled mechanism using a D1 kinetic model; tested predictions against additional experiments and solid-residue characterisation.
- Maintained experimental records, guided research assistants and communicated findings through technical reports and peer-reviewed publication.

SELECTED WATER, ENVIRONMENTAL & R&D PROJECTS

- **Tailings-water treatment, Peru**: investigated metal removal from mine-tailings water by oxidation and selective precipitation; contributed to proposal for Fe, Zn and Mn recovery from Peruvian tailings water.
- **Zinc electrolysis**: planned, set up, ran and reported plate-electrode zinc electrolysis trials end to end, including experimental design, sampling, analysis and recommendations.
- **Rotating-cathode electrowinning**: ran copper and cobalt trials on a continuous rotating-cathode unit, investigating deposit formation and release; proposed for zinc removal from tailings water.
- **Mine-water treatment options**: contributed to a review of treatment options for uranium-bearing mine water.
- **Manganese processing and recovery**: investigated selective precipitation and electrochemical beneficiation routes; supported manganese sulphate monohydrate process development.
- **Portable water-purification bottle (postgraduate design project)**: designed a three-stage unit (pre-screen, hollow-fibre ultrafiltration and activated carbon) rated at 75 L/h.
- **Fe/Al precipitation from leach liquor (academic research)**: investigated MgO, $CaCO_3$ and limestone/eggshell as agents for removal of dissolved iron and aluminium.
- **Phlogopite leaching / potassium recovery (MEng)**: designed nitric-acid leaching experiments, modelled kinetics and characterised feed and residues; research published in *Clays and Clay Minerals* (2022).
- **Additional resource-recovery work**: gold-recovery reactor trials on refinery eluate; lanthanide precipitation experiments; reviewed and edited a Ni-MH battery metal-recovery feasibility study.

ADDITIONAL PROFESSIONAL EXPERIENCE

Business Setup & Marketing Consultant Apr 2025 – Present
Cresco Consulting FZCO, UAE

- Prepare and submit company-registration documentation through the IFZA portal; manage client information and records; maintain the company website and creative content.

Engineer Assistant / Deckhand / Stewardess May 2023 – Apr 2025
Research Vessel Rosa, United States & Bahamas

- Assisted with pumps, plumbing and RO watermaker maintenance and troubleshooting; organised technical drawings, maintenance records and procurement within marine safety procedures.

Academic Journal Editor Nov 2022 – Apr 2023
CACTUS Communications, remote

- Edited scientific manuscripts for technical clarity, consistency and journal requirements; reviewed terminology and provided structured author feedback.

Teaching Assistant (Part-Time) Feb 2019 – Jun 2021
University of Pretoria, South Africa

- Supported laboratory sessions and tutoring in particle technology, kinetics and engineering practice/costing; supervised practical work and assessed student submissions.

Online English Language Consultant Sep 2020 – Mar 2021
MPC Connect, South Africa / remote

- Delivered structured lessons and feedback to non-native English speakers, developing clear communication and independent remote-working skills.

Laboratory Intern (Vacation Work) Dec 2017 – Jan 2019
University of Pretoria, South Africa

- Completed 12 weeks of laboratory placements in mineral leaching and fertiliser production, including experimental optimisation and analytical testing.

EDUCATION AND RESEARCH

MEng Chemical Engineering | University of Pretoria | 2021

BEngHons Chemical Engineering with distinction | University of Pretoria | 2020

BEng Chemical Engineering | University of Pretoria | 2019

- **MEng dissertation:** Development of Kinetic Parameters for Nitric Acid Leaching of Phlogopite and Characterization of the Solid Residue.
- **Publication:** Favel, C.M. & du Plessis, B.J. (2022). Development of Kinetic Parameters for Nitric Acid Leaching of Phlogopite and the Characterization of Solid Products. *Clays and Clay Minerals*, 70, 106–119. doi.org/10.1007/s42860-022-00180-x
- **Environmental Engineering specialisation:** water and effluent treatment, bioreactors, air pollution, waste remediation and environmental regulation.

ResearchGate: researchgate.net/profile/Cheri-Favel

RELEVANT PROFESSIONAL DEVELOPMENT

- **SAMA Return-to-Work Programme, Phase 2** | Selected participant | Oct 2026
- **Project Management: Energy and Water Projects** | PMI / LinkedIn Learning | Feb 2026
- **Occupational Safety and Health: Hazard Communication** | LinkedIn Learning | Feb 2026
- **Approved Engine Course (AEC 1)** | Maritime Professional Training | Feb 2024
- **AI for Project Management: Managing Risk with Generative AI** | NASBA / LinkedIn Learning | Feb 2026
- **LEED Green Associate Exam Preparation** | LinkedIn Learning | Sep 2026 (exam preparation course)
- **STCW Basic Safety Training** | Professional Yachtmaster Training | Dec 2023

LANGUAGES

English (native) | Afrikaans (proficient) | Hebrew (basic)

SCAN OR CLICK TO CONNECT

Contact details, portfolio & research

Opens my portfolio page with project summaries, publication links and contact details.